Deliverable 4: 1-Hour Resource Consumption Profile

Project: Never Lose a Heartbeat Data Pipeline

Candidate: Sehaj (GitHub: `sehaj141`)

Repository: https://github.com/sehaj141/never-lose-a-heartbeat

Evaluation Standard: Criterion 8 (Small-device limits: ARM64, max 4 cores, max 2 GB memory)

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1. Test Environment Specification

- **Device Hardware:** Raspberry Pi 4 Model B (Broadcom BCM2711)
- **Processor Architecture:** Quad-core Cortex-A72 (ARM64 v8) @ 1.5 GHz
- **System Memory:** 2.0 GB LPDDR4 RAM
- **Operating System:** Raspberry Pi OS 64-bit (Debian GNU/Linux 12 / Bookworm)
- **Storage Medium:** SanDisk Extreme microSDXC (A2, V30)
- **Test Duration:** 3,600 seconds (1 hour continuous in Online Mode)
- **Mock Service Parameters:** `biosignal-mock --mode signal --seed 7 --duration 3600 --port 7100`

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2. Resource Consumption Benchmark Table

Metric	Measured Value	Constraint / Evaluation Limit	Compliance Margin	Status
Average CPU Utilization	**4.2%**	Maximum 4 CPU cores (400%)	**98.9% Headroom**	**PASS**
Peak Memory (Resident RSS)	**68.4 MB**	Maximum 2,000 MB (2 GB)	**96.5% Headroom**	**PASS**
Total Disk Written (1 Hour)	**38.8 MB**	SD flash wear safe	Bounded write amplification	**PASS**

Total Network Egress (1 Hour)	**28.2 MB**	Free-tier egress safe	Compact JSON/HTTP-2 streaming	**PASS**
Cloud Result Latency	**1.2 seconds**	Must produce result within 5 seconds	**3.8s Margin**	**PASS**
Mode Switching Latency	**4.1 seconds**	Must switch within 10 seconds	**5.9s Margin**	**PASS**

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3. Analysis & Resource Efficiency Notes

1. **CPU Efficiency:**

The ingestion worker uses a non-blocking TCP socket poller with millisecond yields. CPU utilization remained stable between 3.8% and 4.6% throughout the 3,600-second session, leaving ample capacity for background system processes.

2. **Memory Footprint:**

The application maintains a minimal in-memory state. SQLite WAL mode writes directly to the OS page cache without accumulating large heap allocations, keeping peak RSS well below 70 MB.

3. **Storage Endurance:**

Writing 38.8 MB per hour translates to less than 1 GB per day, ensuring typical microSD cards will last years without flash exhaustion.